UNIT 2 · TERM 1 · WEEKS 4–6

Number

“What follows from what, and how much. Order comes first, and measure is laid on it, as a date is laid on an event.”

The kind of answer: computed · Number on the map

In a child’s words. First put things in order. Only then count them or measure them.

The Institute habit

Chanted drills and constant quizzes, answers memorised and recited. Sticky’s own warning applies: remembering is not thinking.

The book

Arrival on Nomansan Island, the “no rules” that turn out to be rules, the classes, and the first light signals to the mainland.

Week 1 · the kind of answer

Morse code, which Sticky teaches the others. Compute it: with codes of up to four dots and dashes, how many letters can you make? (2 + 4 + 8 + 16 = 30.) Monday puzzle: words a calculator spells upside down.

Week 2 · into the gap

The chess problem from the tests: the starting position with one black pawn moved. Is it possible? It is, if White’s knight went out and came back. The position shows how many moves, not which; the order is hidden under the measure. Number is a gap on the map, so the child writes into it, in Sticky’s voice: why order comes before measure.

Week 3 · the case

Two creatures each the ancestor of the other cannot be: it is a circle, and an order has none. Then the child draws the map of the disciplines from memory, checks that no arrow leads back to where it started (the site’s build refuses one that does), and finds a second order of the twelve units that keeps every arrow.

The workshop · suggested

For a child taking the Nand to Tetris strand, Project 1. Begin with the NAND gate's truth table on paper: two inputs, four rows, and the output is false only when both inputs are true. Then build Not, And and Or out of NAND alone. Every chip the computer will ever have is made of this one.

The end first

Build the numbers as von Neumann did: 0 is an empty box, 1 the box holding 0, 2 the box holding 0 and 1. Each number points to every number before it, so none can be its own ancestor. On the wall map, mark the roots that draw on nothing and the field nothing draws on, then ask: is Worship last because it is least, or because the others are for it?

Fridays

A message sent across the yard at dusk by flashlight in Morse, one child sending and one receiving.

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